Adversarial Vulnerability of Machine-Learning GNSS Spoofing Detectors to Physically Realizable Attacks

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Abstract

Machine-learning detectors are increasingly proposed to protect Global Navigation Satellite System (GNSS) receivers against spoofing, yet their behaviour under a deliberate, adaptive adversary is not well established. This paper reports an adversarial evaluation of fourteen detectors, six deep networks and eight classical models, on receiver observables produced by an open software-defined receiver from the static family of the Texas Spoofing Test Battery, a stationary-receiver setting. Every detector is placed at a common operating point of clean recall 0.95 and is attacked by one shared suite that includes white-box gradient attacks, multi-surrogate transfer, three GNSS domain manipulations and a gradient-free decision-based attack. Each adversarial example is projected onto the physically realizable set so that no reported attack corresponds to a signal the spoofer could not transmit. On clean data no family dominates, with $\boldsymbol{F_1}$ between 0.68 and 0.83. Under the decision-based attack every detector, differentiable or not, is driven to a worst-case spoof recall near zero. The boosted-tree detectors require the smallest perturbation to evade, led by a model that is second only to the best detector on clean data, so resistance to a gradient attack does not by itself indicate robustness. The lone outlier, a support-vector machine, needs roughly six times the perturbation of the most fragile detector to evade, but only because its operating point already calls almost everything spoofed, a robustness-accuracy trade-off rather than a property to design for. GNSS manipulations bounded to a fixed realizable budget leave detection almost unchanged while learned perturbations do not, which places the threat in the learned attack rather than in simple signal shaping. Cross-scenario evaluation shows detection calibrated to 0.95 recall falling to a mean of 0.26 on the unseen overpowered scenario, and cross-satellite failure is specific to a detector and a satellite rather than to a family. Architecture choice therefore trades detection utility against robustness rather than providing a security guarantee, and robustness has to be engineered rather than assumed.

Keywords: GNSS spoofing, Adversarial robustness, Machine learning, Software-defined receiver, Resilient PNT, Decision-based attack

1 Introduction

Global Navigation Satellite Systems provide the positioning, navigation and timing (PNT) reference that underpins power-grid synchronization, financial timestamping, aviation and autonomous platforms (Kaplan and Hegarty 2005). The civil GNSS signal is public and unencrypted, so a spoofer that transmits counterfeit signals can capture a victim receiver and steer its reported position or clock. Field demonstrations have driven a yacht off course through the maritime navigation system (Bhatti and Humphreys 2017) and captured an airborne platform (Kerns et al. 2014), and the underlying detection problem has been studied for more than a decade (Psiaki and Humphreys 2016). Because spoofing attacks the integrity of the PNT solution rather than its availability, a receiver cannot treat the loss of trust as a simple outage, and resilient PNT depends on detecting the attack while it is in progress.

Classical anti-spoofing tests monitor receiver observables such as the carrier-to-noise density ratio, the correlator distortion and the consistency of Doppler and code measurements (Akos 2012; Psiaki and Humphreys 2016). These tests are interpretable and fast, but each targets a specific manipulation and can be evaded by an attacker who respects the monitored quantity. This limitation has motivated a rapid shift toward data-driven detectors that learn the joint signature of an attack across many observables. Recent work applies deep sequence models and classical learners to tracking-stage features and reports high detection accuracy (Borhani-Darian et al. 2024; Li et al. 2025; Chen et al. 2025), and physics-constrained networks now push this line toward generalization across attack conditions (Song et al. 2026). The subfield has expanded quickly, and machine-learning detectors are moving from research prototypes toward integration in receivers.

This progress introduces a risk that GNSS integrity monitoring has not yet confronted. Machine-learning classifiers are vulnerable to adversarial examples, inputs perturbed by a small amount that a human would consider equivalent yet that flip the model decision (Szegedy et al. 2014; Goodfellow et al. 2015). A spoofer is exactly the adaptive adversary that this threat model assumes, since it already controls the transmitted signal and therefore controls, within physical limits, the observables the detector consumes. A detector that is accurate on a fixed attack set can still be defeated by an attacker who shapes the spoofing signal to sit on the benign side of the decision boundary. The relevant question for a fielded detector is therefore not average accuracy but the worst case an adversary can force.

A robustness result is only as strong as the attack used to obtain it, and it is only comparable across detectors that are measured at the same operating point. Two requirements follow, and this study is built on both. The attack has to be matched to the model. A non-differentiable tree ensemble or nearest-neighbour classifier denies a gradient attack a useful gradient, so its survival under such an attack carries no information about its robustness until a gradient-free attack has been applied; the adversarial-machine-learning literature calls this pattern gradient masking and recommends a gradient-free attack before any robustness claim (Athalye et al. 2018; Carlini et al. 2019). The comparison also has to be made at one operating point, because recall, false-alarm rate and F_1 all move with the decision threshold, so detectors read at different thresholds are not directly comparable.

A further requirement is specific to the GNSS setting. An adversarial perturbation in feature space is meaningful only if it corresponds to a signal that a spoofer could actually transmit. Several receiver observables are algebraic functions of others, and perturbing a derived quantity independently of its parent produces a receiver state that cannot physically occur. A credible evaluation must restrict the attacker to the physically realizable set and must respect the couplings among observables, otherwise the reported vulnerability overstates what an attacker can achieve.

Generalization to unseen spoofing conditions is a related and equally practical concern. Physics-constrained detectors that embed the code-carrier relationship as a training constraint have been shown to hold accuracy across held-out scenarios where standard detectors do not (Song et al. 2026). That concerns the clean signal and the distribution shift between attack types. Robustness to a deliberate adversary is a separate axis, and it is the axis this paper measures, alongside cross-scenario and cross-satellite generalization on the same corpus.

We evaluate fourteen detectors on receiver observables generated by the open FGI-GSRx software-defined receiver from the Texas Spoofing Test Battery (TEXBAT) recordings (Humphreys et al. 2012; Humphreys 2016; Finnish Geospatial Research Institute 2022). Processing raw in-phase and quadrature samples through a published receiver, rather than a bespoke pipeline, yields the observables that a real integrity monitor would compute and makes the feature generation reproducible. We restrict every detector to nine physically independent observables, place all fourteen at a common operating point of clean recall 0.95, and attack them with one shared suite (Figure 1). The suite pairs white-box gradient attacks and multi-surrogate transfer with three GNSS domain manipulations and a gradient-free decision-based attack that provides a matched upper bound on the vulnerability of the non-differentiable models. A signal-validity enforcer projects every adversarial example onto the physically realizable set and holds the carrier-to-noise ratio consistent with the correlator power that would produce it.

The evaluation yields four results. At the common operating point no family dominates on clean data, so a comparison drawn at a single threshold does not favour deep or classical models by construction. Under the matched decision-based attack

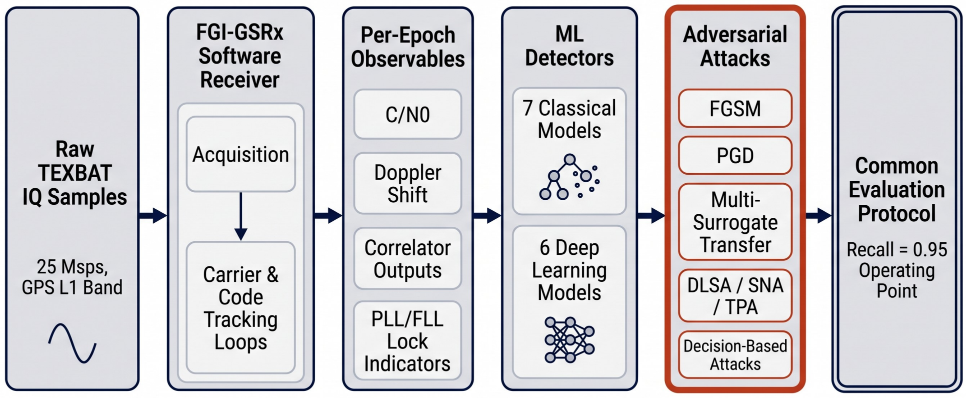

Fig. 1 Signal-level evaluation pipeline. Raw TEXBAT in-phase and quadrature recordings are processed by the FGI-GSRx software-defined receiver into per-epoch observables, which feed eight classical and six deep detectors. Every detector is placed at the common recall-0.95 operating point and attacked by the same suite, from white-box FGSM and PGD through multi-surrogate transfer and the GNSS domain attacks to the gradient-free decision-based attack.

every detector is driven to a worst-case spoof recall near zero, so no architecture in the study provides adversarial robustness, and the apparent robustness of the tree and nearest-neighbour models under transfer attacks is gradient masking. GNSS heuristic manipulations bounded to a fixed budget barely change detection, while learned perturbations defeat it, which locates the threat in the learned attack rather than in naive signal shaping. Cross-scenario evaluation reproduces the generalization collapse, with detection calibrated to 0.95 recall falling to a mean of 0.26 on the unseen overpowered scenario.

The contributions of this paper are the following.

- A model-fair adversarial evaluation protocol for GNSS spoofing detectors that applies one shared attack suite to every model at one common operating point, and that treats non-differentiability as a reason to use a gradient-free attack rather than as evidence of robustness.
- A reproducible signal-level corpus of nine physically independent receiver observables, generated from TEXBAT by an open software-defined receiver, with within-recording labelling that removes the recording-identity confound.
- A signal-validity enforcer that projects adversarial examples onto the physically realizable set and preserves the coupling between reported signal quality and correlator power, so that every reported attack corresponds to a transmittable signal.
- Evidence across fourteen detectors that no architecture is robust under a matched attack, that classical robustness under transfer is gradient masking, and that cross-scenario detection collapses on an unseen attack. Because this holds for every architecture evaluated, the study motivates engineering robustness into the detector rather than selecting it by architecture, and frames the design of such a defence as the central direction for future work rather than a result delivered here.

The remainder of the paper is organized as follows. Section 2 reviews learning-based spoofing detection and adversarial robustness. Section 3 describes the receiver signal model and the observable set. Section 4 states the threat model, and Section 5 details the detectors, the common operating point, the attack suite and the enforcer. Section 6 reports the results, Section 7 discusses their implications for resilient PNT, and Section 8 concludes.

2 Related Work

2.1 Learning-based spoofing detection

Signal-processing anti-spoofing tests operate on quantities the tracking loops already expose. Automatic gain control and received-power monitoring flag the power advantage that many spoofers introduce (Akos 2012), and correlator-domain tests together with checks on the consistency of Doppler and code measurements target the distortion a counterfeit signal leaves on the correlation function (Psiaki and Humphreys 2016). Each test is tied to one physical signature and can be evaded by an attacker who holds that signature within its nominal range, which motivated the move to detectors that learn a joint decision across many observables.

Data-driven detectors now span both deep and classical model families. Among deep models, Borhani-Darian et al. (2024) classify spoofing from cross-ambiguity-function images with a parallelized deep network and report strong detection at moderate to high signal-to-noise ratio, Sung et al. (2022) apply a one-dimensional convolutional network to signal-strength features, Romaniuc et al. (2025) train a long short-term memory model on receiver-independent exchange data and report an F_1 near 0.90, and Yang et al. (2025) combine a bidirectional recurrent network, attention and convolution for a small uncrewed platform and report a weighted F_1 of 0.97. Among classical models, Chen et al. (2022) use a support vector machine, Mahroof et al. (2024) use k-nearest neighbours on the TEXBAT ds3 and ds8 scenarios, and Jullian et al. (2022) compare a multilayer perceptron with a recurrent model. Reported detection accuracy in this line of work is consistently high, which reflects evaluation on a fixed set of attacks with the training and test data drawn from the same conditions.

2.2 Generalization and physics-informed detection

A separate concern is whether a detector trained on one set of attacks still works on attacks it has not seen. Iqbal et al. (2024) motivate a one-class framework by the observation that supervised detectors fail on attack vectors absent from the training distribution, and Chen et al. (2025) address unknown scenarios by augmenting engineered features with a conditional generative model before classification. Li et al. (2025) target real-time time-spoofing detection through a feature-processing pipeline. Song et al. (2026) report that standard learned detectors lose accuracy on unseen spoofing scenarios, and that embedding the code-carrier consistency of authentic signals as a training constraint keeps an accuracy retention rate above 0.92 across ten held-out scenarios, establishing that a physical invariant used as an inductive bias improves generalization across attack conditions on clean signals. Robustness to an adversary that adapts the spoofing signal against the detector is a distinct axis, and it is the axis this study evaluates.

2.3 Adversarial robustness and gradient masking

Machine-learning classifiers can be driven to a wrong decision by a perturbation that is small under the chosen norm (Szegedy et al. 2014). Gradient methods construct such perturbations efficiently, from the single-step sign method (Goodfellow et al. 2015) to iterative projected gradient descent (Madry et al. 2018) and the optimization attack of Carlini and Wagner (2017). Perturbations often transfer between models, so an attacker without access to the target can craft examples on a surrogate (Papernot et al. 2016; Liu et al. 2017). When the target exposes no useful gradient, decision-based attacks reach comparable strength using only the hard output label (Brendel et al. 2018; Chen et al. 2020). This last point is central to a fair comparison across model families. A non-differentiable model such as a tree ensemble or a nearest-neighbour classifier resists gradient attacks by construction, and reading that resistance as robustness is the error that Athalye et al. (2018) name gradient masking. The recommended practice is to attack every model with a method matched to its interface, including a gradient-free attack, before any robustness claim (Carlini et al. 2019).

Adversarial examples have been demonstrated against machine-learning systems that consume physical measurements, including audio (Carlini and Wagner 2018), automotive sensor fusion (Cao et al. 2021), synthetic-aperture radar (Lin et al. 2023) and power-system monitoring (Cheng et al. 2022). In each setting the attacker cannot write arbitrary values into the feature vector and must instead induce them through a physical channel, so a credible evaluation constrains the perturbation to what that channel can produce. The same requirement applies to GNSS, where the spoofer controls the transmitted signal but not the receiver observables directly.

2.4 Adversarial evaluation in the GNSS setting

Adversarial robustness of GNSS spoofing detectors is less explored than detection accuracy. An et al. (2025) demonstrate evasion of a support vector machine operating on reported position, showing that a detector with high nominal accuracy can be defeated by an adversary that shapes its input, and identify the robustness of other detector architectures under black-box conditions as an open question.

The evaluation reported here addresses that question directly. It covers fourteen detectors spanning both families, places them at a shared operating point, applies one attack suite to all of them including a gradient-free decision-based attack that measures vulnerability without relying on gradients, and projects every adversarial example onto the physically realizable set so that each reported attack corresponds to a signal a spoofer could transmit. Table 1 makes the scope difference against that prior study explicit.

Table 1 Scope of this study against the one directly comparable prior evasion study of GNSS spoofing detectors.

	An et al. (2025)	This work
Detectors	one (SVM)	fourteen (8 classical, 6 deep)
Attack domain	reported position	receiver observables
Attacks	DLSA, SNA	FGSM/PGD, transfer, 3 domain, decision-based
Gradient-free attack	no	yes (decision-based boundary)
Realizability	none	signal-validity enforcer
Signal source	CARLA simulation	TEXBAT via FGI-GSRx

3 Receiver Signal Model and Observable Generation

We process the raw Texas Spoofing Test Battery recordings with the FGI-GSRx software-defined receiver (Finnish Geospatial Research Institute 2022), a published open-source receiver, rather than a bespoke pipeline. Each recording is a 25 Msps complex GPS L1 C/A stream. The receiver acquires each satellite and tracks it through standard delay-lock and phase-lock loops. At every integration epoch and for each tracked satellite it emits the observables a real integrity monitor would compute: the carrier-to-noise density ratio C/N_0, its running mean, the noise-floor estimate, the Doppler and carrier-frequency estimates, the prompt in-phase and quadrature correlator accumulators I_P and Q_P, the delay-lock discriminator and the phase- and frequency-lock indicators. Processing raw samples through a published receiver, rather than a hand-built feature extractor, yields the observables a fielded receiver would actually see and makes the feature generation reproducible. The receiver acquires the authentic satellites cleanly: Figure 2 shows the acquisition metric separating the acquired satellites from the noise floor and a single sharp correlation peak for the strongest satellite.

A spoofer controls the transmitted signal amplitude $A_s(t)$, its carrier phase $\theta_s(t)$ and its code phase $\tau_s(t)$. These map deterministically to the receiver observables. Amplitude sets the prompt power $I_P^2 + Q_P^2$ and hence C/N_0, carrier phase sets the split between I_P and Q_P, and code phase sets the delay-lock discriminator and the pseudorange. The ds7 scenario is a concrete instance. It is a power-matched, carrier-phase-aligned takeover in which θ_s is ramped so that the composite in-phase and quadrature amplitude stays near constant while the tracking loops are captured, after which the relative code phase drifts and induces a growing clock offset (Humphreys 2016). Figure 3 shows this signature on one satellite. After the takeover the C/N_0 of the spoofed recording settles about two decibels below the authentic reference, while the Doppler stays matched and the delay-lock discriminator stays near zero, which is the stealthy behaviour a data-driven detector is asked to catch.

A multi-correlator view of the same takeover shows how little it disturbs the shape of the correlation function. Figure 4 plots the correlation magnitude of PRN 23 at seventeen taps across plus or minus two chips, for an authentic interval and a spoofed interval. The peak stays at zero offset and the two functions almost coincide, departing

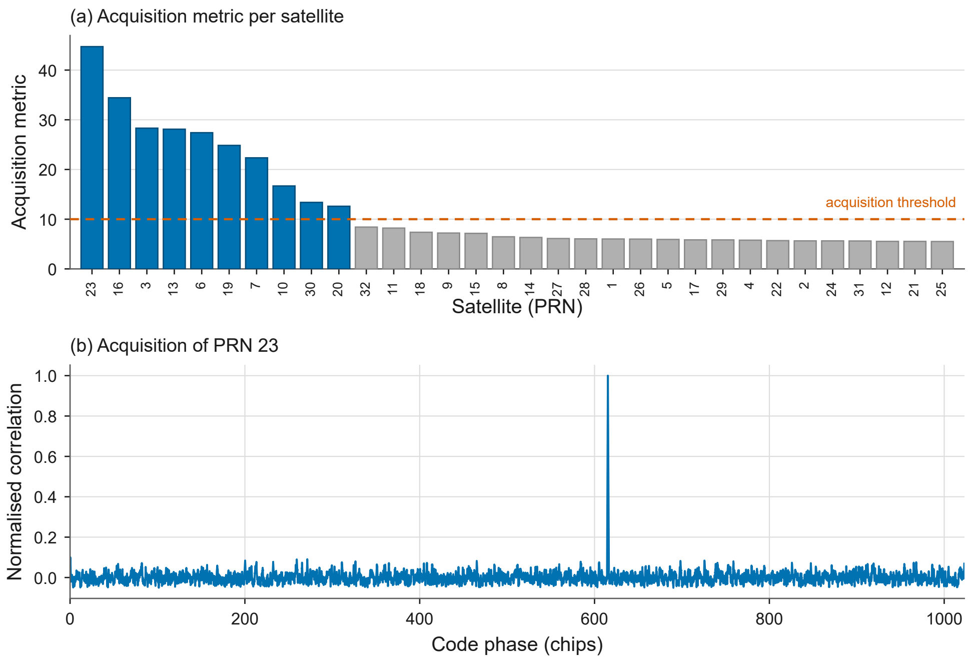

Fig. 2 Receiver validation on the clean static recording. (a) Acquisition metric per satellite; ten satellites are acquired above the threshold and the rest fall in the noise floor. (b) The code-phase correlation for the strongest satellite shows a single sharp acquisition peak.

by at most about two percent of the peak. A test on correlation shape alone therefore has little to detect here, which is consistent with the power-matched, carrier-phase-aligned design of the scenario and is part of why a detector that combines the correlator state with the other observables is preferred.

3.1 Physically independent observables

Several receiver observables are algebraic functions of others, so perturbing a derived quantity on its own produces a receiver state that cannot physically occur. On our corpus the carrier-frequency column equals the Doppler column exactly, the prompt power equals $I_P^2 + Q_P^2$ exactly, and the prompt phase equals $\operatorname{atan2}(Q_P, I_P)$ exactly; the frequency-lock filter output is defined only during pull-in and is absent in steady-state tracking. We therefore restrict every detector to the nine physically independent observables, representing the correlator state by the raw accumulators I_P and Q_P rather than their derived power and phase. The one remaining soft coupling, C/N_0 as a function of the correlator power, is enforced during attack generation (Section 5). A feature-space attack that moved a derived observable independently of its parents would correspond to a signal no transmitter could produce, so this restriction is what keeps the reported attacks physically meaningful.

Figure 5 shows the distribution of the observables for genuine and spoofed epochs pooled over the corpus. The separation is carried mainly by C/N_0 and the correlator

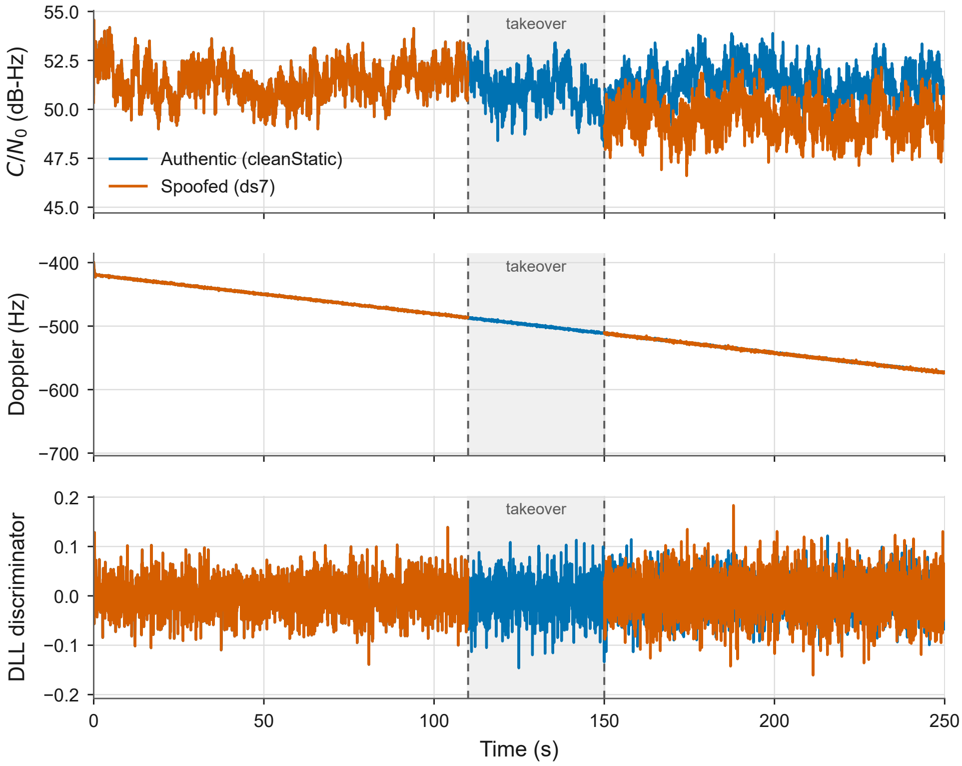

Fig. 3 Spoofing-signature time history for one satellite (PRN 23): the authentic clean static recording against the ds7 spoofed recording. After the takeover the carrier-to-noise ratio settles about two decibels below the authentic reference, while the Doppler stays matched and the delay-lock discriminator stays near zero, the stealthy signature of the power-matched, carrier-phase-aligned takeover. The shaded 110 to 150 s transition band is excluded from the corpus, so only the authentic recording is drawn across it.

accumulators, while the Doppler is clustered by satellite and is weakly discriminative on its own, which is consistent with the stealthy, power-matched nature of the strongest scenarios. Pooled C/N_0 is slightly higher for genuine than for spoofed epochs, which at first looks opposite to the power advantage a spoofer needs to capture the tracking loops. The two scenarios with the most spoofed epochs, ds3 and ds7, are deliberately power-matched rather than overpowered, and ds7 settles below the authentic level after capture (Section 3); only the overpowered ds2 scenario shows spoofed C/N_0 above genuine, and it contributes fewer epochs. The pooled direction is therefore a property of the scenario mix, dominated by the stealthy scenarios, rather than a general rule that spoofing raises C/N_0.

4 Threat Model

We evaluate every detector under one common threat model. We specify it in the standard adversarial-machine-learning terms that separate the attacker's goal from its

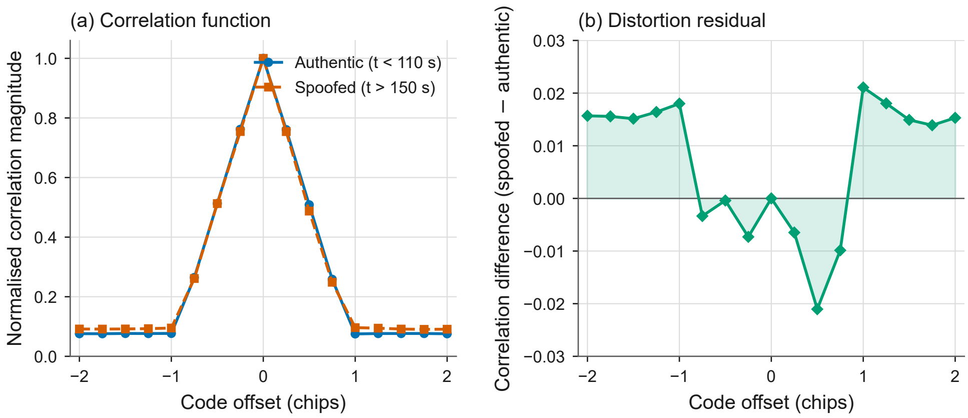

Fig. 4 Multi-correlator correlation function of PRN 23 under ds7, at seventeen taps across plus or minus two chips. (a) Correlation magnitude for an authentic interval ($t < 110$ s) and a spoofed interval ($t > 150$ s), each normalised to its peak; the two nearly coincide. (b) The spoofed minus authentic residual: the peak stays at zero offset while the spoofer raises the off-peak correlator floor and slightly suppresses the late shoulder, by at most about two percent of the peak. The small distortion is consistent with the power-matched, carrier-phase-aligned design of the takeover.

knowledge of the detector and its capability to shape the input (Biggio and Roli 2018). The attacker is a spoofer that already transmits counterfeit signals and seeks to convert a spoofed observation the detector would flag into one it accepts. A detector maps an observable vector $\mathbf{x} \in \mathbb{R}^d$ to a spoof score $s(\mathbf{x}) \in [0, 1]$ and flags the epoch as spoofed when $s(\mathbf{x}) \geq \eta$ at an operating threshold η (Section 5.3). The attacker manipulates the transmitted signal, which in turn moves the receiver observables within the physical limits set out in Section 3, seeking an additive perturbation $\boldsymbol{\delta}$ that turns a spoofed observation the detector would flag ($y = 1$, $s(\mathbf{x}) \geq \eta$) into one it accepts,

$$s(\mathbf{x} + \boldsymbol{\delta}) < \eta, \qquad \|\boldsymbol{\delta}\|_\infty \leq \varepsilon, \qquad \mathbf{x} + \boldsymbol{\delta} \in \mathcal{F}, \tag{1}$$

where $\|\cdot\|_\infty$ is measured in the min-max normalised observable space, ε is the perturbation budget, and $\mathcal{F}$ is the physically realizable set imposed by the signal-validity enforcer of Section 5.5. The gradient attacks (FGSM, PGD) and their transfer variants fix $\varepsilon \in \{0.05, 0.10, 0.20\}$; the GNSS domain attacks apply a fixed budget in each detector's own input space. The gradient-free decision-based attack is the exception: rather than test a fixed budget it searches for the smallest perturbation that flips each sample, measured as a min-max normalised L_∞ distance, and this per-sample minimum is the fragility measure reported in Section 6. The min-max fragility measure maps directly to physical units. The carrier-to-noise density ratio spans about 57 dB-Hz across the training corpus, so the most fragile detector, evaded at a median distance near 0.02, falls to a carrier-to-noise change of at most about 1.1 dB, while the most robust needs about 0.11, roughly 6 dB. In the standardised space the fixed gradient budget of 0.10 corresponds to about 0.5 dB on the same observable or about 200 Hz of Doppler. The perturbations that defeat the detectors are therefore physically small changes a spoofer can impose rather than abstract normalised numbers.

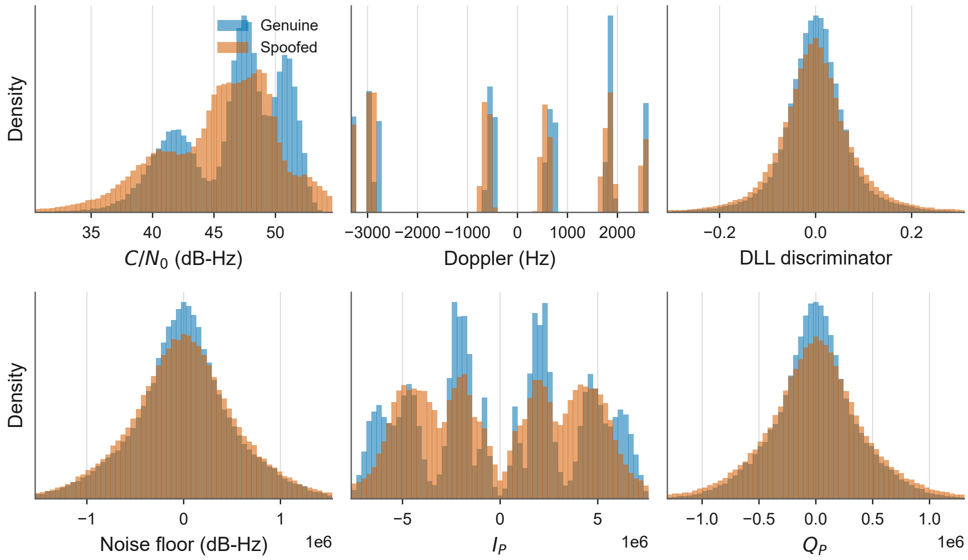

Fig. 5 Distribution of the receiver observables for genuine and spoofed epochs, pooled over the corpus. The class separation is carried mainly by the carrier-to-noise ratio and the correlator accumulators.

Every adversarial example is projected onto the physically realizable set by the signal-validity enforcer of Section 5, including the coupling between reported signal quality and correlator power, so no reported attack corresponds to a signal the spoofer could not transmit.

The attacker's access to the detector depends on the detector. For the differentiable deep detectors the attacker has white-box gradient access. This is a deliberate worst-case assumption: a fielded receiver more likely faces a query-limited or transfer-only adversary, so the white-box result upper-bounds the deep detectors' vulnerability. For the non-differentiable tree and nearest-neighbour detectors the attacker has hard-label access, meaning it observes only whether the target receiver continues to accept its signal. A spoofer obtains this hard-label signal in practice by watching whether the victim maintains lock and reports the intended solution. We treat the non-differentiability of the tree and nearest-neighbour detectors as a reason to use a gradient-free attack rather than as evidence of robustness, because a model that merely denies useful gradients can still be evaded through its decision boundary (Athalye et al. 2018; Carlini et al. 2019). The decision-based boundary attack, which uses only hard labels, provides the matched upper bound on the vulnerability of these detectors (Brendel et al. 2018).

5 Methods

5.1 Corpus construction

We use the static family of the Texas Spoofing Test Battery (Humphreys et al. 2012): the clean static recording as the authentic baseline and the ds2, ds3 and ds7 spoofing scenarios. ds2 is an overpowered time push at about 10 dB advantage, ds3 is a matched-power time push at about 1.3 dB, and ds7 is a power-matched, carrier-phase-aligned takeover (Humphreys 2016). The two dynamic scenarios are excluded so that a static-versus-dynamic motion difference cannot confound the comparison. Each recording is processed by the receiver of Section 3 into per-epoch observables.

Labelling is done within each recording rather than by recording identity. In a spoofed recording the epochs before the documented takeover carry the authentic label and the epochs after the pull-off carry the spoofed label, with the transition band discarded; the clean static recording is authentic throughout. Drawing both classes from the same recordings removes the recording-identity shortcut, under which a detector could separate the classes by learning which file an epoch came from rather than the spoofing signature. The resulting corpus holds about 209,000 labelled epochs, close to balanced (about 51% spoofed), so no resampling is applied.

Train, validation and test partitions are formed by a leakage-free block-temporal split. Within each satellite and recording segment the epochs are ordered by time and cut into contiguous blocks of 70%, 10% and 20%, with a purge gap dropped at each boundary. A random per-epoch split would leak, because adjacent epochs of the same satellite are near-duplicates through temporal autocorrelation, so a random split lets almost-identical epochs fall on both sides of the partition and inflates the reported performance.

5.2 Detectors

We evaluate fourteen detectors. The eight classical models are a random forest (Breiman 2001), gradient boosting (Friedman 2001), XGBoost (Chen and Guestrin 2016), LightGBM (Ke et al. 2017), a k-nearest-neighbour classifier, a multilayer perceptron, a decision tree and a support-vector machine with a radial-basis-function kernel, tuned by the same five-fold grid search as every other classical model. The six deep models are a one-dimensional convolutional network, a long short-term memory network (Hochreiter and Schmidhuber 1997), a bidirectional LSTM, a convolutional-recurrent hybrid, a Transformer (Vaswani et al. 2017) and a temporal convolutional network (Bai et al. 2018). The multilayer perceptron is a shallow feed-forward model and is grouped with the classical detectors. Each model consumes the nine observables of Section 3.1. The fourteen span the classifier families used across the cited detection literature of Section 2: tree ensembles and boosting, instance-based classification, kernel methods, a shallow feed-forward network, and, among the deep models, the convolutional, recurrent, hybrid and attention-based branches of sequence modelling.

A linear-kernel support-vector machine was also trained on this corpus and separates the classes far worse than every detector in Table 2, so it is excluded as uninformative: a detector that does not separate the classes to begin with is not a meaningful adversarial-robustness test. The support-vector machine that An et al. (2025) attack instead uses a radial-basis-function kernel; that is the kernel trained here, reaching a clean F_1 of 0.682, weaker than every other detector but well above chance, so it is retained as the fourteenth detector and the SVM family An et al. attack is represented under the present protocol.

5.3 Common operating point

A detector's recall, false-alarm rate and F_1 all depend on its decision threshold, so a comparison drawn at different thresholds confounds the detector with its operating point. We fix one common operating point for every model,

$$\eta^\star = \max\{\eta : R_{\mathrm{val}}(\eta) \geq 0.95\}, \tag{2}$$

the highest threshold whose recall $R_{\mathrm{val}}(\eta)$ on the clean validation block still meets a floor of 0.95, which is the safety-first choice for an integrity monitor. All clean and adversarial metrics are reported at $\eta^\star$. Because the false-alarm rate at this recall floor is informative in its own right, we also report a second operating point defined by a clean-validation false-alarm ceiling of 0.05, so that a high false-alarm rate at the recall floor is visible rather than hidden by the choice of threshold.

5.4 Attack suite

Every detector is attacked by one shared suite at $\eta^\star$. The white-box attacks are the single-step fast gradient sign method (Goodfellow et al. 2015) and iterative projected gradient descent (Madry et al. 2018), both bounded by the L_∞ budget ε and applied to the differentiable deep detectors. For the non-differentiable classical detectors the gradient attacks are transferred from deep surrogates, both from a single convolutional surrogate and from a mean-logit ensemble of three surrogates, since an ensemble of surrogates transfers more strongly than one (Papernot et al. 2016; Liu et al. 2017).

The suite also includes three GNSS domain attacks. We adapt the data location shift attack and the similarity noise attack of An et al. (2025), which were defined in the reported-position domain against a single support vector machine, to the receiver-observable domain, and we add a temporal pattern attack that modulates the Doppler and C/N_0 observables. These attacks are heuristic and physically motivated rather than optimised against the detector.

The decisive element of the suite is a gradient-free decision-based boundary attack applied to every detector. For each correctly detected spoofed sample it searches, in the min-max normalised observable space, for the smallest L_∞ perturbation that moves the sample across the decision boundary to the authentic side at $\eta^\star$,

$$\rho(\mathbf{x}) = \min_{\boldsymbol{\delta}} \|\boldsymbol{\delta}\|_\infty \quad \text{subject to} \quad s\big(\Pi_{\mathcal{F}}(\mathbf{x}+\boldsymbol{\delta})\big) < \eta^\star, \tag{3}$$

the minimal-perturbation objective of Szegedy et al. (2014) solved with hard-label queries in the manner of Brendel et al. (2018), evaluated on realizable examples through the enforcer $\Pi_{\mathcal{F}}$ of Section 5.5. The per-sample minimum $\rho(\mathbf{x})$ is the fragility measure, and a detector's fragility is the median of ρ over its correctly detected spoofed samples: a smaller median means a smaller perturbation suffices, so the detector is more fragile. Because the attack uses only the hard decision, it applies uniformly to differentiable and non-differentiable detectors and provides the matched measure of vulnerability that a transferred gradient attack cannot. The suite characterises L_∞ vulnerability; other norms such as L_2 or L_0 (Carlini and Wagner 2017) may expose different weaknesses and are outside the scope of this study.

5.5 Signal-validity enforcer

Every adversarial example, at both training and evaluation, is projected onto the set $\mathcal{F}$ of physically attainable receiver states, $\tilde{\mathbf{x}} = \Pi_{\mathcal{F}}(\mathbf{x} + \boldsymbol{\delta})$, by a signal-validity enforcer fitted on the training partition. Each observable is first clipped to a conservative data-driven band. The enforcer then couples the reported C/N_0 to the correlator power: it fits the log-linear relation $\widehat{C/N_0} = a + b \log_{10}(I_P^2 + Q_P^2)$ by least squares on the training partition and admits only reported values within the empirical residual band of that prediction,

$$q_1 \leq (C/N_0) - \big(a + b \log_{10}(I_P^2 + Q_P^2)\big) \leq q_{99}, \tag{4}$$

where q_1 and q_{99} are the first and ninety-ninth percentiles of the training residuals. An attacker therefore cannot raise the reported signal quality without also supplying the correlator power that would produce it. This encodes the physical relation between the spoofer's amplitude control and the reported observables and is what separates a realizable attack from an arbitrary feature perturbation. Figure 6 illustrates the effect. An attack that raises the reported C/N_0 without supplying the correlator power that would produce it leaves the physically attainable band, and the enforcer projects it back onto that band, so the enforcer is the mechanism by which every attack in this study is kept transmittable. The enforcer acts per epoch, so it guarantees that each epoch is individually realizable but does not constrain the epoch-to-epoch continuity of the amplitude, phase and code trajectories; a temporal-smoothness constraint on that trajectory is a natural extension left to future work.

5.6 Evaluation protocol and statistics

Detection is measured under three views. The primary view is the block-temporal split within the mixed corpus. The cross-scenario view holds out one spoofing scenario entirely and trains on the remainder, which measures generalization to an unseen attack type. The leave-satellite view holds out one satellite, which measures generalization across satellites. Every fold sets the common operating point on its own validation block and retrains each detector on its own training block, so no information from the held-out fold enters training.

The effect of the fixed-budget attacks is summarised by the attack success rate, the fraction of initially correct detections that an attack flips at $\eta^\star$. Uncertainty on the clean F_1 is reported as a 10,000-sample bootstrap confidence interval, and the median minimum L_∞ perturbation of each detector is reported with a 10,000-sample

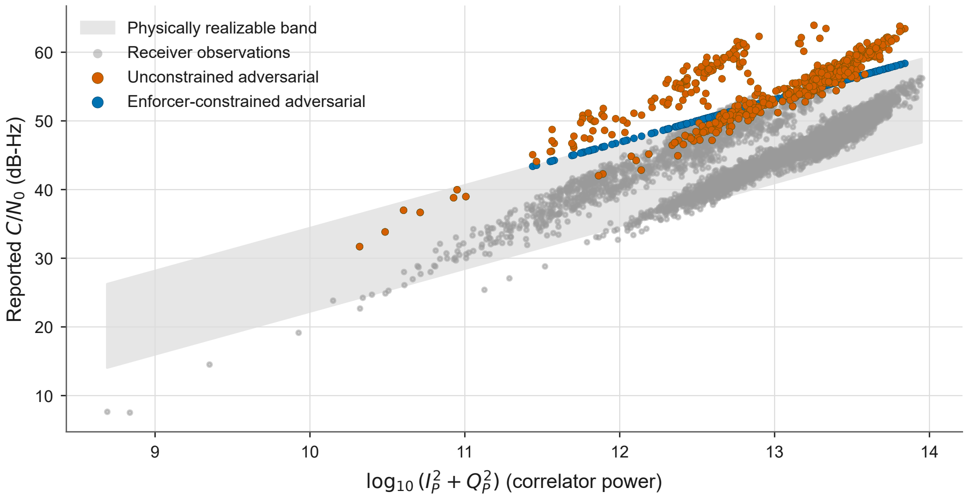

Fig. 6 The signal-validity enforcer on the C/N_0 to correlator-power manifold. Receiver observations lie in the physically realizable band. An adversarial attack that raises the reported C/N_0 without supplying the correlator power leaves the band (unconstrained), and the enforcer projects it back onto the band, so the reported attack corresponds to a transmittable signal.

bootstrap interval over the attacked samples. Pairwise detector differences, computed on the clean test-set decisions, are tested with the mid-p McNemar test (Dietterich 1998) under Holm correction across all pairs, and heterogeneity across attacks is tested with the Friedman test (Demšar 2006).

6 Results

6.1 Clean detection

At the common operating point no architecture family dominates on clean data. Table 2 and Figure 7 report detection at the recall-0.95 threshold. The F_1 scores fall in a band from 0.682 to 0.833, and the classical and deep detectors are interleaved through it: the best detector is the classical random forest at 0.833, the deep temporal convolutional network at 0.805 sits above five classical models, and the weakest detector is classical, the radial-basis-function support-vector machine at 0.682, the only detector below 0.70. The area under the ROC curve tells a similar story, ranging from 0.648 to 0.908 with no clean separation by family. The bootstrap confidence intervals on F_1 are narrow, within about ± 0.005, so the ordering is stable but the spread across families is small. A comparison drawn at a single operating point therefore does not, by construction, favour deep or classical detectors.

The false-alarm rate at the recall floor is high, from 0.34 to 0.96, reflecting both the difficulty of the corpus once the recording-identity shortcut is removed and, at the upper end, the weak discriminative power of the least separable detector, the support-vector machine (Section 5.2). Reporting the second operating point makes this explicit rather than hidden. When a clean-validation false-alarm ceiling of 0.05

Table 2 Clean detection at the common operating point (clean-validation recall = 0.95). Recall, false-alarm rate (FAR), F_1 and AUC are on the test partition; $\eta^\star$ is the decision threshold. No family dominates.

Model	Family	$\eta^\star$	Recall	FAR	F_1	AUC
RandomForest	classical	0.705	0.941	0.337	0.833	0.876
GradientBoosting	classical	0.660	0.937	0.350	0.827	0.876
XGBoost	classical	0.625	0.972	0.451	0.811	0.885
LightGBM	classical	0.505	0.966	0.464	0.804	0.863
KNN	classical	0.090	0.957	0.462	0.800	0.884
DecisionTree	classical	0.070	0.967	0.522	0.787	0.803
MLP	classical	0.175	0.947	0.587	0.758	0.866
SVM	classical	0.510	0.986	0.960	0.682	0.648
TCN	deep	0.210	0.944	0.427	0.805	0.908
BiLSTM	deep	0.095	0.955	0.662	0.741	0.869
LSTM	deep	0.155	0.952	0.679	0.735	0.843
CNN-LSTM	deep	0.080	0.958	0.693	0.734	0.855
CNN-1D	deep	0.115	0.964	0.737	0.725	0.856
Transformer	deep	0.225	0.956	0.794	0.707	0.817

is imposed instead, two of the tree-based detectors, XGBoost and the decision tree, cannot reach it at any threshold, and the gradient-boosting model that meets it on validation does not carry the low false-alarm rate to the test set. Detection on this corpus is thus honest but far from solved, which is the right baseline against which to measure adversarial degradation.

6.2 Adversarial robustness

Under the matched attack no detector is robust. Against the gradient-free decision-based attack every one of the fourteen detectors is driven to a worst-case spoof recall near zero, so at an unbounded budget an attacker flips every correctly detected spoofed sample to the authentic side. The informative quantity is therefore how large a perturbation the attack needs, shown in Figure 8 as the median minimum L_∞ perturbation that evades each detector.

This measure overturns the reading that non-differentiable detectors are robust, for most of the classical family. The five tree-based classical detectors, gradient boosting, XGBoost, LightGBM, the decision tree and the random forest, form the most fragile cluster, a chain of overlapping 95% bootstrap intervals spanning a median of 0.019 to 0.025; gradient boosting has the smallest point estimate but its interval overlaps XGBoost's, so the point-estimate ordering within the cluster is not resolved. The multilayer perceptron (0.039) and the temporal convolutional network (0.041) sit above this cluster and overlap each other but not it. The nearest-neighbour detector and five of the six deep models, the Transformer, CNN-1D, BiLSTM, LSTM and CNN-LSTM, form a second chain of overlapping intervals from 0.055 to 0.067, so the nearest-neighbour detector is not the clean classical exception in the deep band that its point estimate alone would suggest.

The support-vector machine stands apart from every other detector: a median perturbation of 0.111 (0.105 to 0.117) evades it, roughly 1.7 times the perturbation needed for the next-most-robust detector and roughly six times that needed for the most fragile, and its interval does not overlap any other detector's. Section 7 interprets why this coincides with the weakest clean performance in Table 2 rather than the strongest.

The apparent robustness of the tree and nearest-neighbour detectors under a transferred gradient attack is thus an artifact of the attack, not a property of the architecture: when the same detectors are attacked through their decision boundary they are among the easiest to evade. The multi-surrogate transfer is stronger than the single-surrogate transfer, yet both understate the vulnerability that the decision-based attack reveals.

The three GNSS domain attacks, each bounded to a fixed budget in the detector's own input space and projected onto the realizable set, are far weaker than the learned attacks. Their attack success rate has median 0.076, interquartile range 0.031 to 0.198 and a worst case of 0.548, well below the learned attacks' median success rate of 0.35 (FGSM) and 0.76 (PGD) at matched budgets, so naive in-budget manipulation of the observables is markedly less effective than the learned perturbations. The threat therefore lies in the learned attack rather than in simple signal shaping, and every learned attack in this study corresponds to a transmittable signal because of the enforcer of Section 5.5. Attack generation is fast relative to inference for all detectors, so the attack is not excluded by cost.

6.3 Generalization

Generalization to unseen conditions fails, and it fails by detector and by condition rather than by family. Figure 9(a) reports cross-scenario recall when a spoofing scenario is held out of training. Averaged over the three held-out scenarios the mean recall is 0.53 for the classical detectors and 0.49 for the deep detectors, with a standard deviation across the held-out scenarios of 0.20 and 0.30 respectively, and the overpowered ds2 scenario is the worst case for every detector, with the random forest falling to a recall of 0.08. This reproduces, on an independent receiver pipeline, the collapse toward chance that physics-constrained work has reported for learned spoofing detectors on unseen scenarios (Song et al. 2026). The behaviour is scenario-dependent rather than a fixed family ranking: on the carrier-phase-aligned ds7 scenario two deep detectors reach recalls of 0.94 and 0.89, above every classical detector's best of 0.86.

Figure 9(b) reports recall across the eleven held-out satellites. Most satellites are handled well by most detectors, with per-detector median recall between 0.56 and 1.00, but the worst held-out satellite drives specific detectors near zero: the gradient-boosting model falls to a recall of 0.007 on one held-out satellite, joined on the same fold by two other boosted-tree detectors while the random forest and decision tree are unaffected, and the one-dimensional convolutional network falls to 0.060 on a different satellite while every other deep detector holds up. With eleven satellites this

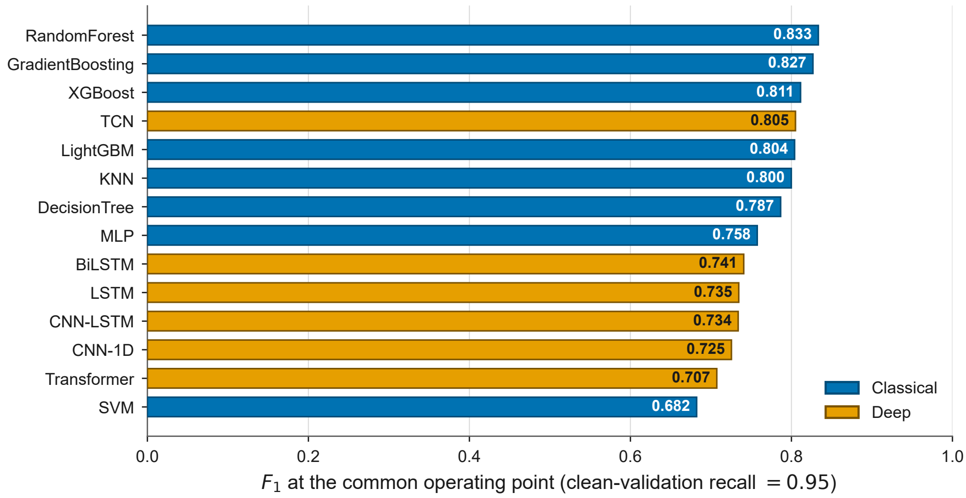

Fig. 7 Clean detection F_1 at the common operating point, by detector and family. The scores span a narrow band and the classical and deep detectors are interleaved, so no family dominates.

is a per-fold observation rather than a hypothesis-tested pattern, but the direction is consistent: cross-satellite failure is per-architecture and not predicted by the family label, not even at the finer grain of boosted trees versus other tree ensembles.

6.4 Statistical significance

The detectors are pairwise distinguishable. Across the McNemar tests over all detector pairs, with Holm correction, 88 of 91 pairs differ at the adjusted 0.05 level, so the comparison is not undermined by detectors that make identical errors. The Friedman test finds strong heterogeneity across attacks within each family, over five attack conditions (deep $\chi^2(4) = 20.1$, $p = 4.7 \times 10^{-4}$; classical $\chi^2(4) = 26.8$, $p = 2.2 \times 10^{-5}$), which confirms that attack choice, and in particular the decision-based attack, is what drives the degradation.

7 Discussion

The results carry one message for resilient positioning, navigation and timing: a machine-learning spoofing detector cannot be assumed robust because it is accurate, and its architecture does not decide the question. On clean data no family dominates, and under a matched attack every family is defeated. Architecture choice buys a trade-off between detection utility and robustness, not a security guarantee. The trade-off is direct: Figure 10 ranks the detectors by clean F_1 and by the perturbation needed to evade them, and the two orderings are close to reversed, so the detector that is strongest on clean data is among the weakest under attack. The support-vector machine is the clearest instance, since it resists the decision-based attack more than any other detector yet has the weakest clean F_1 and the highest false-alarm rate, so its robustness is bought at a utility cost that a fielded integrity monitor could not accept.

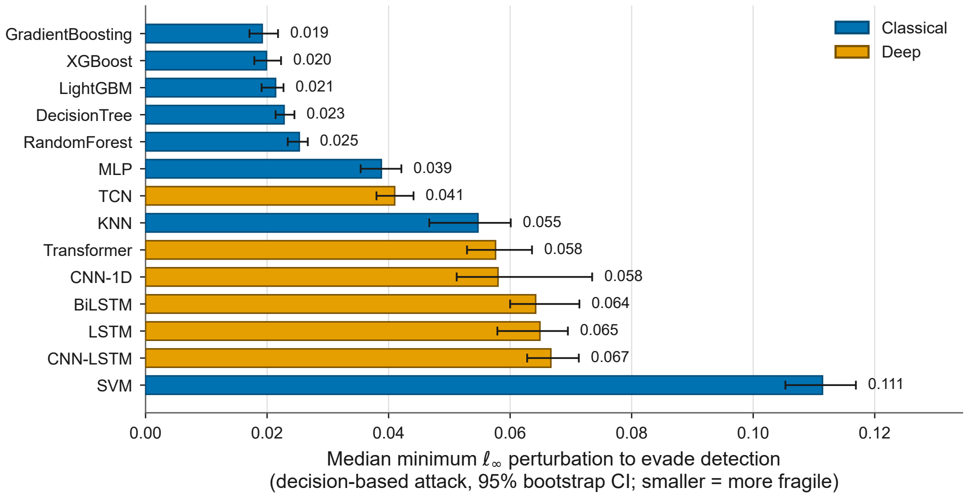

Fig. 8 Median minimum L_∞ perturbation needed to evade each detector under the gradient-free decision-based attack, with 95% bootstrap confidence intervals over the 500 attacked samples per detector. Smaller is more fragile. The five tree-ensemble classical detectors are the easiest to evade; the support-vector machine is the single robust outlier, needing roughly six times the perturbation of the most fragile detector. Within the tree cluster and within the nearest-neighbour/deep-model band the intervals overlap, so neither internal ranking is resolved.

The mechanism is visible at its operating point: it labels 98.6% of spoofed epochs and 96.0% of genuine epochs as spoofed (Table 2), so before any attack its decision surface already assigns almost everything to the spoofed side; an adversary has to cross more of that surface to reach the small region the model still calls genuine, which is exactly what a larger measured perturbation reflects. This is a robustness-accuracy trade-off in the sense of Tsipras et al. (2019) rather than a property to design for: the same operating point that inflates the false-alarm rate also inflates the measured robustness, so the fragility ranking on its own is not a ranking of which detector an operator should prefer. Figure 11 places all four axes side by side, and no detector scores well on all of them: strength on clean detection, on the false-alarm rate, on adversarial robustness and on cross-scenario recall are in tension rather than aligned.

The finding that matters most for how these detectors are evaluated is that non-differentiability is not robustness. A tree ensemble or a nearest-neighbour classifier denies a gradient attack a useful gradient, so its survival under such an attack carries no information about its robustness. Measured through the decision boundary instead, the ordering reverses: the boosted-tree detectors are the easiest to evade, and the gradient-boosting model, second only to the random forest on clean data, is the most fragile of all. Any future evaluation of a GNSS spoofing detector that claims robustness for a non-differentiable model should include a gradient-free attack for this reason.

The physical-realizability constraint is what makes the threat credible rather than a feature-space artifact. Because every adversarial example is projected onto the attainable receiver states and the coupling between reported signal quality and correlator

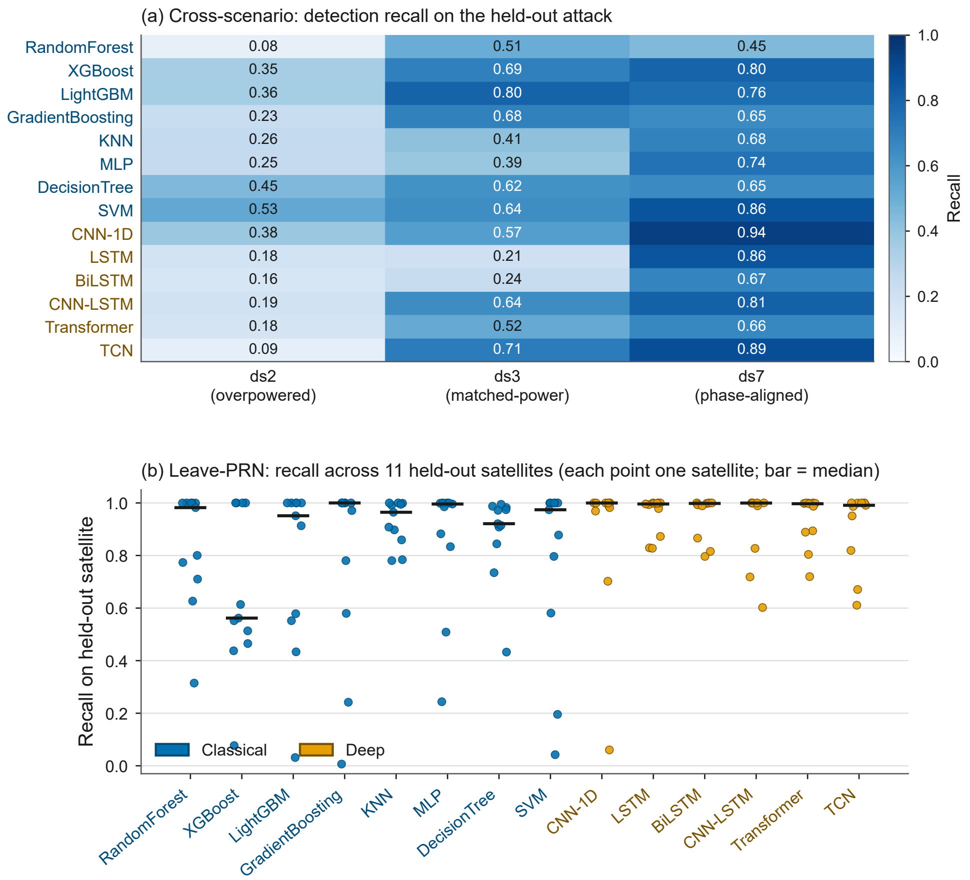

Fig. 9 Generalization. (a) Cross-scenario detection recall when a spoofing scenario is held out of training; the overpowered ds2 scenario collapses every detector. (b) Recall across held-out satellites, one point per satellite, bar at the median; specific satellites break specific detectors while others of the same family hold up on the same fold. Label colour marks the family.

power is enforced, each reported attack corresponds to a signal a spoofer could transmit. Within that constraint the GNSS domain attacks are almost harmless, while the learned perturbations succeed, which locates the threat in the learned attack rather than in simple signal shaping and shows that the realizability constraint does not by itself protect the detector.

Generalization behaves the same way as robustness. Detection calibrated on the mixed corpus collapses on an unseen overpowered attack and breaks unpredictably on individual satellites, and which condition breaks which detector is not predicted by the family label. That the collapse reproduces, on a separate receiver pipeline, the generalization loss that Song et al. (2026) report for learned detectors on unseen scenarios indicates that it is a property of learned spoofing detection rather than of one pipeline.

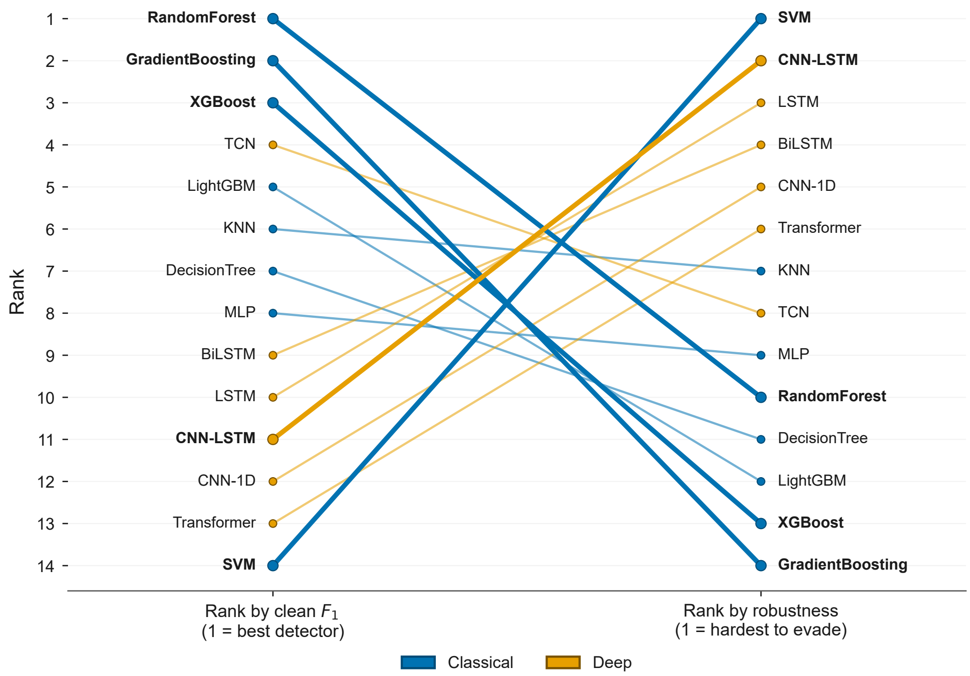

Fig. 10 Detector rank by clean F_1 (left) against rank by the median L_∞ perturbation needed to evade the decision-based attack (right). The two orderings are close to reversed: the detectors strongest on clean data are the easiest to evade, and the support-vector machine moves from weakest on clean data to hardest to evade.

Designing a GNSS spoofing detector with robustness that holds under a matched, physically realizable attack is the natural next step. Empirical adversarial training and certified approaches such as randomized smoothing are the candidate directions, and evaluating whether either restores robustness at an acceptable false-alarm rate is the focus of ongoing work. The evidence here sets the requirement that such a defence must meet: it has to withstand a gradient-free attacker at a common operating point, not merely a transferred gradient attack.

Framed in integrity terms, the consequence is direct. Figure 12 places every detector on the missed-detection against false-alarm plane, where missed detection is the integrity risk that precedes hazardously misleading information and the false-alarm rate is the continuity risk. On clean data every detector sits in a narrow band of low missed detection, so each looks like a serviceable monitor. A single decision-based attack held to a fixed, physically small budget of $\varepsilon = 0.10$, about 6 dB of C/N_0, lifts every detector out of that band: missed detection rises to between 0.44 and 1.00, so the monitor accepts most or all of the spoofed epochs it would otherwise flag. The ordering of the lifted points reproduces the fragility ranking of Section 6.2 on a safety-relevant axis, and no detector stays in a region an integrity monitor could use.

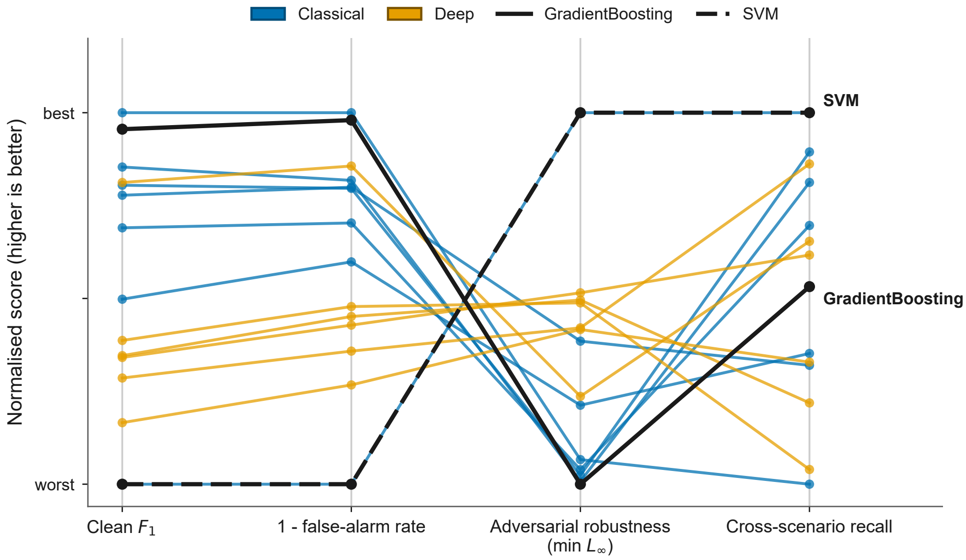

Fig. 11 Each detector across four axes, normalised so higher is better on every axis: clean F_1, one minus the false-alarm rate, adversarial robustness and cross-scenario recall. No detector scores well on all four. The gradient-boosting model is second only to the random forest on clean detection and worst on robustness, and the support-vector machine is the reverse: worst on clean detection, best on robustness.

Two findings compound for a practitioner before any adversary is considered. The clean false-alarm rate at the safety-first operating point is high, from 0.34 to 0.96 (Table 2), and cross-scenario recall collapses on an unseen attack. On this evidence none of the fourteen detectors is deployable as a stand-alone, single-epoch integrity monitor today, independently of adversarial robustness. A fielded monitor would fuse decisions over time, through a vote or a sequential test across epochs, which trades detection latency for a lower per-decision false-alarm rate; the per-epoch numbers reported here are the input to such a scheme, not the operating performance of a deployed system.

The evaluation covers the static family of the benchmark, which is GPS L1 C/A, with three held-out spoofing scenarios and the satellites present in those recordings. The gradient-masking result and the fragility ordering depend on the detectors and the attack rather than on the specific signal, which leads us to hypothesise that they would persist on other civil signals, such as BeiDou B1I or Galileo E1, and on multi-frequency receivers, with the realizability enforcer recalibrated to the per-signal C/N_0-to-power relation. This is a hypothesis rather than a tested result, since the present evidence is confined to GPS L1 C/A, and confirming the carry-over would require repeating the protocol on those signals. Extending the protocol to a wider set of scenarios, to multi-constellation and multi-frequency observables, and to dynamic platforms is a natural direction for further work and would test whether the conclusions hold across a broader operating envelope.

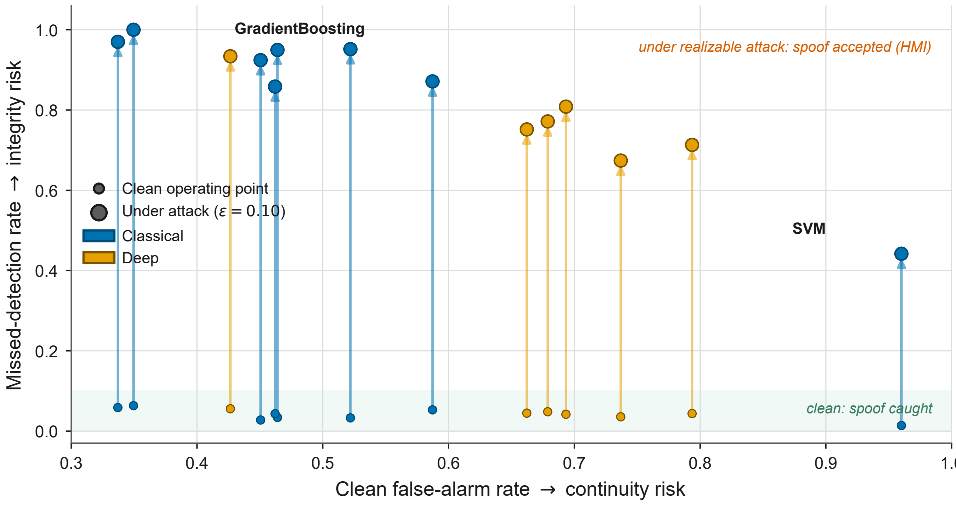

Fig. 12 Operational integrity view. Each detector is shown on the missed-detection against false-alarm plane at its clean operating point (small marker) and after a single decision-based attack at a fixed, physically small budget $\varepsilon = 0.10$ (about 6 dB of C/N_0; large marker), with the false-alarm rate held at its clean value because the evasion attack targets spoofed epochs. Missed detection is the integrity risk that precedes hazardously misleading information; the false-alarm rate is the continuity risk. Every detector leaves the low-missed-detection band under the attack, and none remains in a region an integrity monitor could use.

8 Conclusion

We reported a model-fair adversarial evaluation of fourteen machine-learning GNSS spoofing detectors on receiver observables produced by an open software-defined receiver from the Texas Spoofing Test Battery. Every detector was placed at a common operating point and attacked by one shared suite that included a gradient-free decision-based attack, and every adversarial example was projected onto the physically realizable set so that no reported attack corresponds to an untransmittable signal.

Four results follow. On clean data no architecture family dominates, with F_1 between 0.68 and 0.83. Under the decision-based attack every detector is driven to a worst-case spoof recall near zero, and measured through the decision boundary the boosted-tree detectors are the easiest to evade, led by a model that is second only to the best detector on clean data, so resistance to a gradient attack does not by itself indicate robustness. The lone outlier, a support-vector machine, needs roughly six times the perturbation of the most fragile detector to evade, but only because its operating point already calls almost everything spoofed, a robustness-accuracy trade-off rather than a property to design for. GNSS manipulations bounded to the same realizable budget are almost harmless, so the threat is the learned attack rather than naive signal shaping. Cross-scenario and cross-satellite generalization fails per detector and per condition, reproducing on an independent pipeline the generalization loss reported for learned spoofing detection on unseen scenarios.

Taken together, these results argue that robustness has to be engineered into a GNSS spoofing detector rather than assumed from its accuracy or its architecture, and that a credible robustness claim requires a gradient-free attack at a common operating point against physically realizable perturbations. Building and certifying a detector that meets that requirement is the subject of ongoing work.

Data and code availability. The derived receiver-observable corpus and the evaluation and figure-generation code are available from the corresponding author on reasonable request. The raw Texas Spoofing Test Battery recordings are distributed by the University of Texas at Austin Radionavigation Laboratory, and the FGI-GSRx software-defined receiver is released under the GPLv3 licence.

Competing interests. The authors declare no competing interests.

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